

W6 - 1.5 Solving Quadratic Equations – Part 2: Solve Using the Q.F.

MCR3U

1) Use the discriminant to determine the number of roots for each quadratic equation.

a) $x^2 - 10x + 25 = 0$

b) $3x^2 + 4x + \frac{4}{3} = 0$

c) $2x^2 - 8x + 9 = 0$

d) $-2x^2 + 0.75x + 5 = 0$

2) Solve each quadratic using the quadratic formula. Give exact answers.

a) $x^2 + 4x - 3$

b) $-x^2 + 12 = 9x$

c) $x^2 = -5x + 2$

d) $x^2 - 3x + 1 = 6$

e) $x^2 + 6x + 9 = 0$

f) $4x^2 - 6x - 1 = 0$

g) $5x^2 - 3x - 1 = 0$

h) $-x^2 + 7x - 18 = 0$

i) $4x^2 - 25 = 0$

j) $3x^2 - 7x - 4 = x^2 - 4x$

k) $8x^2 + 4x - 5 = 0$

l) $4x^2 - 18x = 0$

3) Solve each quadratic equation using any method

a) $3x^2 - 12x = 0$

b) $2x^2 + 4x - 6 = 0$

c) $3x^2 + 5x - 2 = 0$

d) $4x^2 - 11x - 8 = 0$

4) Three lengths of pipe measuring 24 cm, 31 cm, and 38 cm will be used to create a right triangle. The same length of pipe will be cut off each of the three pipes to allow a right triangle to be created. What is that length?